HelixAI Microservice System Architecture Specification

1. Overview

HelixAI is a production-grade FastAPI microservice tailored for Data & AI Engineering workloads.
It features autonomous tool calling, local open LLM inference (Llama 3 8B, Mistral 7B), JWT authentication,
multi-role RBAC (Admin, Analyst, Data Engineer, User), and sliding-window rate limiting.

2. Data Engineering Tool Matrix

- Autonomous SQL AI Agent: Converts natural language questions into dialect-aware SQL (PostgreSQL, Snowflake, BigQuery)
- AI Document & Web Researcher: Fuses uploaded CSV files, PDFs, or architectural text documents with real-time web search
- Schema Guard: Validates Pydantic v2 schemas and JSON structural integrity for microservice APIs.

3. Performance Benchmarks

- Sub-50ms token authentication and authorization verification.
- Dual-quota sliding window rate limits: 20 Requests Per Minute (RPM) and 10,000 Tokens Per Minute (TPM).
- 100% containerized deployment with multi-stage Docker build and automated CI via GitHub Actions.